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Econometrics II

5. Time Series Models (Part 1)

Regression with autocorrelated errors

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Key questions and objectives

This chapter focuses on the following key questions and objectives:

- What are consequences of autocorrelation for the OLS estimator?
- What is the efficient estimator in the presence of autocorrelation?
- How can we test and correct for autocorrelation?

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5.6.1 Re-specification of the model





- The transformation $\epsilon_t - \rho\epsilon_{t-1}$ will generate homoskedastic non-autocorrelated errors because ν_t satisfies the Gauss–Markov conditions.
- The transformed model is given by (Cochrane–Orcutt transformation)

- If ρ is known, estimating 5.4 by OLS produces an **approximate GLS estimator**. The estimator is only approximate because the transformation cannot be applied to the first observation (because y_0 and $\mathbf{x}_0$ are not observed).
- The transformation of the first observation is

Note that $Var(\sqrt{1 - \rho^2}\epsilon_1) = \sigma_v^2$.

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- $$\hat{\rho} = \left(\sum_{t=2}^T \mathbf{e}_{t-1}^2 \right)^{-1} \left(\sum_{t=2}^T \mathbf{e}_t \mathbf{e}_{t-1} \right) \quad (5.6)$$

- Using $\hat{\rho}$ instead of ρ produces the FGLS estimator, which is asymptotically equivalent to the GLS estimator.
- As an alternative, the **iterative Cochrane–Orcutt procedure** estimates ρ and β recursively until convergence is achieved:

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Testing for first-order autocorrelation

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Testing for first-order autocorrelation

5.3 Testing for first-order autocorrelation

5.3.1 Asymptotic tests

(1) *t*-test

- We may use the OLS residuals from equation 5.1 to detect serial correlation in the error term. In particular, we consider the **auxiliary regression** of the OLS residual e_t upon its lag e_{t-1} (with or without an intercept term).
- This auxiliary regression produces an estimate for ρ and its standard error. In the absence of lagged dependent variables in equation 5.1, the corresponding *t*-test is asymptotically valid.

(2) Breusch-Godfrey Lagrange multiplier test

- The Breusch-Godfrey test is based on the R^2 of the auxiliary regression (including an intercept term). The **test statistic**, $(T - 1)R^2$, has a χ^2 distribution with one degree of freedom.
- The test can be extended to higher orders of autocorrelation by including additional lags of the residual and adjusting the degrees of freedom accordingly.

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Testing for first-order autocorrelation

- **Example:** the auxiliary regression for a test of third-order autocorrelation is

$$e_t = \alpha + \rho_1 e_{t-1} + \rho_2 e_{t-2} + \rho_3 e_{t-3} + \nu_t, \quad t = 4, 5, \dots, T.$$

- If the model of interest includes a lagged dependent variable (or other explanatory variables that are correlated with lagged error terms), then the regressors $\mathbf{x}_t$ should be included in the auxiliary regression. This makes sure that the test statistics have the appropriate approximate distribution.

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Testing for first-order autocorrelation

5.3.2 The Durbin–Watson test

- The Durbin-Watson test can be applied in **small samples**.
- **Assumptions:**
 - The $\mathbf{x}_t$ s are deterministic. In this case, all error terms are independent of all explanatory variables. Note that this excludes the inclusion of lagged dependent variables in the model.
 - The vector $\mathbf{x}_t$ contains an intercept term.
- **Durbin–Watson test statistic:**

$$dw = \frac{\sum_{t=2}^T (e_t - e_{t-1})^2}{\sum_{t=1}^T e_t^2} \approx 2 - 2\hat{\rho}.$$

- If $dw \approx 2$, then the first-order autocorrelation coefficient $\rho \approx 0$.
- If $dw \ll 2$, then there is evidence for positive autocorrelation ($\rho > 0$).
- If $dw \gg 2$, then there is evidence for negative autocorrelation ($\rho < 0$).

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Testing for first-order autocorrelation

- **Critical values** cannot be tabulated for general use because, even under the null, the distribution of dw depends not only upon the sample size T and the number of variables K in $\mathbf{x}_t$, but also upon the actual values of the $\mathbf{x}_t$ s.
- However, it is possible to compute **upper and lower limits for the critical values** of dw , denoted d_L and d_U respectively, that depend only upon T and K . The true critical value d_{crit} is between the bounds that are tabulated: $d_L < d_{\text{crit}} < d_U$.

Number of observations	Number of regressors (incl. intercept)							
	$K = 3$		$K = 5$		$K = 7$		$K = 9$	
	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U
$T = 25$	1.206	1.550	1.038	1.767	0.868	2.012	0.702	2.280
$T = 50$	1.462	1.628	1.378	1.721	1.291	1.822	1.201	1.930
$T = 75$	1.571	1.680	1.515	1.739	1.458	1.801	1.399	1.867
$T = 100$	1.634	1.715	1.592	1.758	1.550	1.803	1.506	1.850
$T = 200$	1.748	1.789	1.728	1.810	1.707	1.831	1.686	1.852

Source: Verbeek (2017), Table 4.8

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Testing for first-order autocorrelation

- **Rejection rule** for a one-sided test against positive autocorrelation ($\rho > 0$):
 - ① If $dw < d_L$, then we reject the null hypothesis.
 - ② If $dw > d_U$, then we do not reject the null hypothesis.
 - ③ If $d_L < dw < d_U$, then we are unable to accept or reject the null hypothesis (**inconclusive region**).
- When the alternative hypothesis is the presence of negative autocorrelation ($\rho < 0$), the true critical value is between $4 - d_U$ and $4 - d_L$, so that no additional tables are required.
- The existence of an inconclusive region and the requirement that the Gauss–Markov conditions, including normality of the error terms, are satisfied are **drawbacks** of the Durbin–Watson test.
- In contrast, asymptotic tests are approximately valid, even without normal error terms, and can be extended to allow for lagged dependent variables in $\mathbf{x}_t$.

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Example: demand for ice cream

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Example: demand for ice cream

5.4 Example: demand for ice cream

- **Research question:** what explains the demand for ice cream?
- **Data:** time series data with 30 four-weekly observations from 18 March 1951 to 11 July 1953. The variables are:

cons:	consumption of ice cream per head (in pints)
income:	average family income per week (in US Dollars)
price:	price of ice cream (per pint)
temp:	average temperature (in Fahrenheit)
time:	index from 1 to 30

- The time series patterns of consumption, price and temperature suggest that the temperature is an important determinant for the consumption of ice cream.

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Example: [demand for ice cream](#)

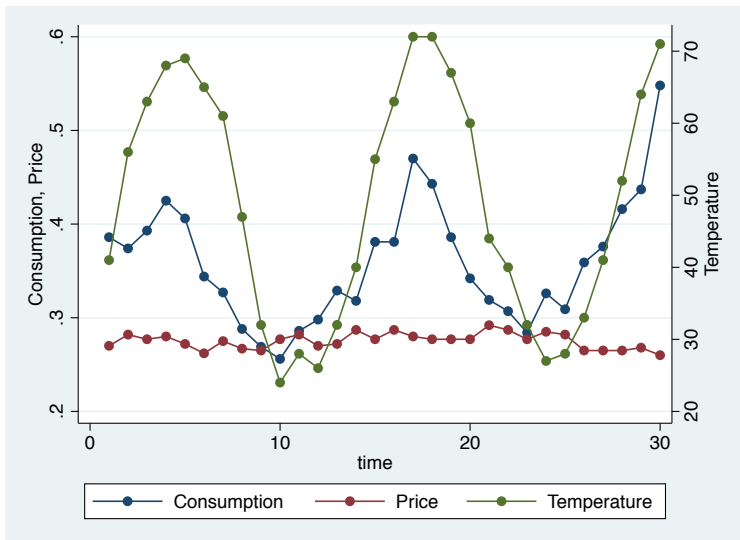
- Load packages and data in R:

```
> library(tidyverse)
> library(dynlm)
> library(lmtest)
> library(orcutt)
> data("icecream")
> glimpse(icecream, width=50)
Rows: 30
Columns: 4
$ price <dbl> 0.270, 0.282, 0.277, 0.280, 0.27...
$ cons  <dbl> 0.386, 0.374, 0.393, 0.425, 0.40...
$ income <dbl> 78, 79, 81, 80, 76, 78, 82, 79, ...
$ temp  <dbl> 41, 56, 63, 68, 69, 65, 61, 47, ...
> tsdata <- ts( icecream, start=1 )
```

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Example: demand for ice cream

- Time series patterns of consumption, price and temperature:



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Example: demand for ice cream

• Estimation results from an OLS regression

```
> fit <- dynlm( cons ~ price + income + temp, data = tsdata )
> summary( fit )
```

Time series regression with "ts" data:

Start = 1, End = 30

Call:

```
dynlm(formula = cons ~ price + income + temp, data = tsdata)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.1973151	0.2702162	0.730	0.47179
price	-1.0444140	0.8343573	-1.252	0.22180
income	0.0033078	0.0011714	2.824	0.00899 **
temp	0.0034584	0.0004455	7.762	3.1e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.03683 on 26 degrees of freedom

Multiple R-squared: 0.719, Adjusted R-squared: 0.6866

F-statistic: 22.17 on 3 and 26 DF, p-value: 2.451e-07

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Example: demand for ice cream

- **Durbin-Watson test** for positive autocorrelation

```
> dwtest( fit )
```

Durbin-Watson test

```
data: cons ~ price + income + temp
```

```
DW = 1.0212, p-value = 0.0003024
```

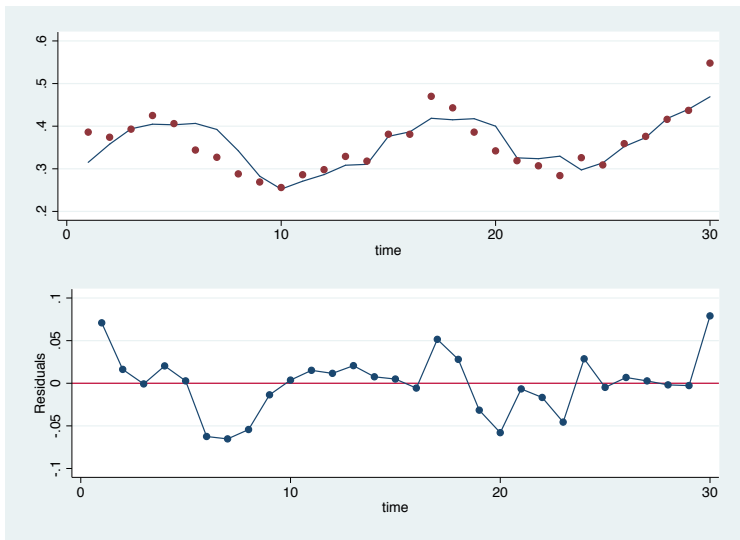
```
alternative hypothesis: true autocorrelation is greater than 0
```

- The upper and lower limits for the critical value at the 5% significance level are $d_L = 1.21$ and $d_U = 1.65$, respectively. Hence, we clearly reject the null hypothesis against the alternative of positive autocorrelation.
- A **graphical inspection** of residuals confirms that positive (negative) values for the error term are more likely to be followed by positive (negative) values. We conclude that the inclusion of the variable temp in the model is insufficient to capture the seasonal fluctuation in ice cream consumption.

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Example: demand for ice cream

- Graph:



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Example: demand for ice cream

- Next, we estimate the first-order autocorrelation coefficient with a regression of e_t on e_{t-1}

```
> coeftest( dynlm( residual ~ -1 + L(residual) ) )
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
L(residual)	0.40063	0.17742	2.2581	0.03192 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- The asymptotic t -test clearly indicates rejection of $H_0 : \rho = 0$ against first-order autocorrelation.

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Example: demand for ice cream

- The **Breusch-Godfrey test** uses the R^2 from an auxiliary regression including the explanatory variables and an intercept term.

```
> bgaux <- dynlm( residual ~ L(residual) + price + income + temp,
+               data = tsdata )
> summary( bgaux )$r.squared
[1] 0.1622135
```

- The **test statistic** is $(T - 1)R^2 = 29 \cdot 0.1622 = 4.70$.
- We reject the null hypothesis of no autocorrelation because the empirical value of the test statistic is larger than the 5% critical value of 3.84 of a χ^2 distribution with one degree of freedom.

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Example: demand for ice cream

- The **iterative Cochrane–Orcutt method** yields the results below:

```
> summary( cochrane.orcutt(fit) )
```

Call:

```
dynlm(formula = cons ~ price + income + temp, data = tsdata)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.15714771	0.28962931	0.543	0.59222
price	-0.89239565	0.81085041	-1.101	0.28157
income	0.00320274	0.00154606	2.072	0.04878 *
temp	0.00355839	0.00055468	6.415	1.022e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.0319 on 25 degrees of freedom

Multiple R-squared: 0.6489 , Adjusted R-squared: 0.6068

F-statistic: 15.4 on 3 and 25 DF, p-value: < 7.015e-06

Durbin-Watson statistic

(original): 1.02117 , p-value: 3.024e-04

(transformed): 1.54884 , p-value: 5.061e-02

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Example: demand for ice cream

- The finding of autocorrelation may indicate misspecification of the dynamics. Therefore, we change the specification of the model and include one or more lagged variables:

```
> fit.ltemp <- dynlm( cons ~ price + income + temp + L(temp),
+                      data = tsdata )
>
> summary(fit.ltemp)
```

Time series regression with "ts" data:

Start = 2, End = 30

Call:

```
dynlm(formula = cons ~ price + income + temp + L(temp), data = tsdata)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.1894822	0.2323170	0.816	0.42274
price	-0.8383023	0.6880209	-1.218	0.23490
income	0.0028673	0.0010533	2.722	0.01189 *
temp	0.0053321	0.0006704	7.953	3.5e-08 ***
L(temp)	-0.0022039	0.0007307	-3.016	0.00597 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.02987 on 24 degrees of freedom

Multiple R-squared: 0.8285, Adjusted R-squared: 0.7999

F-statistic: 28.98 on 4 and 24 DF, p-value: 7.1e-09

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Example: [demand for ice cream](#)

- The Durbin–Watson test statistic

```
> dwtest(fit.ltemp)
```

Durbin-Watson test

```
data: fit.ltemp
```

```
DW = 1.5822, p-value = 0.02876
```

```
alternative hypothesis: true autocorrelation is greater than 0
```

is in the inconclusive region, as $d_L = 1.14$ and $d_U = 1.74$. The value is close to the upper bound. Hence, we may choose not to reject the null of no autocorrelation.

- The lagged temperature has a significant negative effect on ice cream consumption while the current temperature has a positive effect. This may indicate an increase of demand when the temperature rises.

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Moving average errors

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Moving average errors

5.5 Moving average errors

- A moving average error process models a form of autocorrelation where only particular error terms are correlated while all others have a zero correlation.
- Moving average structures often arise when the sampling interval is smaller than the interval for which the variables are defined.
- **Example:**
 - **Research question:** what is the relationship between the **yearly** change in prices (inflation) and economic background variables, such as money supply?
 - **Data:** prices changes, y_t , and background variables, $\mathbf{x}_t$, are observed **half-yearly**.
 - **Model for half-yearly data** is given by

$$y_t = \mathbf{x}_t' \boldsymbol{\beta} + \nu_t, \quad t = 1, 2, \dots, T \text{ (half-yearly)}$$

where error term ν_t has variance σ_ν^2 and satisfies the Gauss-Markov assumptions.

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Moving average errors

- **Example (cont.):**
 - **Model for yearly data** is given by

$$y_t + y_{t-1} = (\mathbf{x}_t + \mathbf{x}_{t-1})' \boldsymbol{\beta} + \nu_t + \nu_{t-1}, \quad (5.7)$$

$$y_t^* = \mathbf{x}_t^{*'} \boldsymbol{\beta} + \epsilon_t. \quad (5.8)$$

- The **covariance matrix of the error term vector** is

$$E[\boldsymbol{\epsilon}\boldsymbol{\epsilon}'] = \begin{pmatrix} 2\sigma_\nu^2 & \sigma_\nu^2 & 0 & \cdots & 0 \\ \sigma_\nu^2 & 2\sigma_\nu^2 & \sigma_\nu^2 & \cdots & 0 \\ 0 & \sigma_\nu^2 & 2\sigma_\nu^2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 2\sigma_\nu^2 \end{pmatrix}. \quad (5.9)$$

- Note that the covariance matrix contains a large number of zeros.
- The structure in 5.9 is called a **first order moving average process**.

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Moving average errors

- A general first order moving average process is

$$\epsilon_t = \nu_t + \alpha\nu_{t-1}, \quad (5.10)$$

where $|\alpha| < 1$.

- Estimation of regression models with moving average errors is more complicated than with autoregressive errors because the transformation generating “Gauss–Markov errors” is complicated. If appropriate software is lacking, one could apply OLS with standard errors that are corrected for the presence of autocorrelation.

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[Solutions to autocorrelation](#)

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Solutions to autocorrelation

5.6 Solutions to autocorrelation

5.6.1 Re-specification of the model

- Finding of autocorrelation may indicate that the **model is misspecified**. If this is the case, we may change the model.
- Misspecification may arise from **dynamic misspecification**, **omitted variables** and **functional form misspecification**.
- Suppose that the relationship between y_t and $\ln x_t$ is

$$y_t = \beta_1 + \beta_2 \ln x_t + \epsilon_t,$$

where we assume that x_t increases with t .

- If we estimate the model

$$y_t = \beta_1 + \beta_2 x_t + \epsilon_t,$$

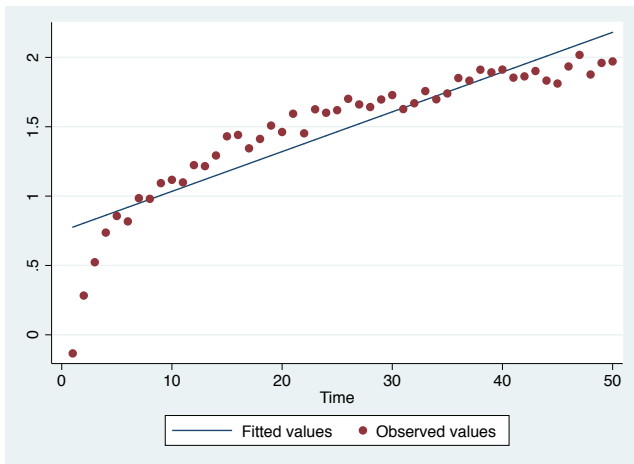
we have autocorrelated error due to misspecification, as indicated by a graph of observed values and fitted values (see below).

- The **solution** in this case is not to use FGLS but to **change the functional form** and include $\ln x_t$ rather than x_t .

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[Solutions to autocorrelation](#)

- **Example:** misspecification of functional form produces autocorrelation



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Solutions to autocorrelation

- Autocorrelation may also arise from an incorrect **dynamic specification**.
- Suppose the **static model**

$$y_t = \mathbf{x}'_t \boldsymbol{\beta} + \epsilon_t, \quad (5.11)$$

where the error term ϵ_t follows a first-order autoregressive process $\epsilon_t = \rho\epsilon_{t-1} + \nu_t$. The model describes $E[y_t|\mathbf{x}_t]$.

- As an alternative model, we may consider a **dynamic model** that includes $\mathbf{x}_{t-1}$ and y_{t-1} :

$$E[y_t|\mathbf{x}_t, \mathbf{x}_{t-1}, y_{t-1}] = \mathbf{x}'_t \boldsymbol{\beta} + \rho(y_{t-1} - \mathbf{x}'_{t-1} \boldsymbol{\beta}). \quad (5.12)$$

- The dynamic regression model is

$$y_t = \mathbf{x}'_t \boldsymbol{\beta} + \rho y_{t-1} - \rho \mathbf{x}'_{t-1} \boldsymbol{\beta} + \nu_t, \quad (5.13)$$

where ν_t does not exhibit autocorrelation. Hence, including a lagged dependent variable and lagged exogenous variables gives a specification that does not suffer from autocorrelation.

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Solutions to autocorrelation

5.6.2 Heteroskedasticity-and-autocorrelation-consistent standard errors for OLS

- Consider the model with an autoregressive error term ϵ_t :

$$y_t = \mathbf{x}_t' \boldsymbol{\beta} + \epsilon_t.$$

- We may apply OLS and **adjust standard errors**.
- If $E[\mathbf{x}_t \epsilon_t] = \mathbf{0}$ and $E[\epsilon_t \epsilon_{t-s}] = 0$ for $s = H, H+1, \dots$, the OLS estimator is consistent and its covariance matrix can be estimated by

$$\widehat{Var^*}(\mathbf{b}) = \left(\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right)^{-1} TS^* \left(\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right)^{-1}, \quad (5.14)$$

where

$$S^* = \frac{1}{T} \sum_{t=1}^T e_t^2 \mathbf{x}_t \mathbf{x}_t' + \frac{1}{T} \sum_{j=1}^{H-1} w_j \sum_{s=j+1}^T e_s e_{s-j} (\mathbf{x}_s \mathbf{x}_{s-j}' + \mathbf{x}_{s-j} \mathbf{x}_s'). \quad (5.15)$$

Note that we obtain the White covariance matrix if $w_j = 0$.

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Solutions to autocorrelation

- It is common to use weights that decrease linearly with j as $w_j = 1 - j/H$. This is compatible with the idea that the impact of the autocorrelation of order j diminishes with j .
- Standard errors computed from 5.14 are referred to as **heteroskedasticity-and-autocorrelation-consistent (HAC) standard errors** or simply **Newey–West standard errors**.
- Derivation of 5.14 can be found in Newey and West (1987).
- The R function `vcovHAC()` from the package `sandwich` computes HAC standard errors.

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[Key terms](#)

Key terms

- autocorrelation
- Breusch-Godfrey test
- Cochrane-Orcutt estimators
- Durbin-Watson test
- dynamic misspecification
- dynamic model
- feasible generalized least squares (FGLS)
- first-order autocorrelation
- generalized least squares (GLS)
- HAC standard errors
- iterative Cochrane-Orcutt procedure
- moving average process
- negative autocorrelation
- Newey–West standard errors
- positive autocorrelation
- Prais-Winsten estimator
- static model
- stationary process
- time series data

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[References](#)

References

This chapter is based on Verbeek (2017), Chapter 4.

Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708.

Verbeek, M. (2017). *A Guide to Modern Econometrics*. John Wiley & Sons, Chichester.